

VISTA-MED: Integrating AI-Based MRI Segmentation with Virtual Reality for Medical Education

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Abstract

Background

Innovations in Artificial Intelligence (AI) and Virtual Reality (VR) are driving transformative changes in medical imaging and education by enabling automated segmentation and immersive visualization. Traditional anatomy and imaging instruction often lack spatial interactivity, limiting comprehension and diagnostic reasoning. This study introduces VISTA-MED, an integrated AI-VR platform designed to enhance medical imaging education through real-time MRI visualization and segmentation.

Methods

The system employs the nnU-Net architecture for automated segmentation of brain tumor regions using the BRATS2020 dataset. Preprocessing included bias field correction, image registration to the MNI152 template, and noise reduction. Segmentation outputs were converted into 3D-compatible formats and integrated into a Unity-based VR environment supporting interactive exploration of anatomical structures. Performance was evaluated using Dice coefficient, Intersection over Union (IoU), and precision-recall analysis.

Results

The segmentation model achieved a mean Dice score of 0.914, indicating high overlap between predicted masks and ground truth annotations. IoU values were consistent with Dice scores, confirming accurate delineation of tumor boundaries. Internal testing validated smooth rendering and stable frame rates in the VR environment, enabling real-time interaction without noticeable latency. Qualitative assessment confirmed anatomical fidelity of AI-generated masks.

Conclusion

VISTA-MED demonstrates the technical feasibility of combining AI-driven segmentation with immersive VR visualization for medical imaging education. This integration enhances spatial understanding and diagnostic reasoning, addressing limitations of traditional anatomy instruction. Future work will focus on usability studies, scalability, and educational impact assessment to support broader adoption in clinical and academic settings.

Background

Innovations in AI and VR are redefining medical education by offering innovative approaches to teaching and clinical training. VR creates immersive, interactive environments that enhance engagement, knowledge retention, and procedural skill development among medical students [1, 2, 3]. It provides cost-

effective, standardized simulations of anatomy and clinical procedures, addressing limitations of traditional methods that rely on static 2D images [4]. In parallel, AI-driven platforms like 3D Slicer enable automated image analysis and segmentation, equipping learners with essential competencies for interpreting complex medical data [5, 6]. These technologies have demonstrated particular value in diagnostic fields such as radiology and pathology, improving precision and enabling personalized learning experiences [7].

Traditional anatomy education often struggles to convey spatial relationships using 2D representations. Research consistently shows that 3D visualization significantly improves comprehension and retention compared to conventional approaches [8, 9]. Similarly, VR-based learning enhances anatomical knowledge, spatial reasoning, and clinical decision-making, offering safe alternatives to cadaver-based training and remote learning opportunities [10, 11, 12]. Beyond anatomy, VR has been shown to improve diagnostic reasoning and treatment planning, while maintaining comparable accuracy to traditional imaging methods [13, 14].

The convergence of AI and VR presents a unique opportunity to advance medical education by combining automated segmentation with immersive visualization. AI-powered tools can transform 2D medical images into interactive 3D models, while VR provides an intuitive platform for real-time exploration and feedback [15, 16]. The combination of AI and VR in medicine spans various applications, including visualization enhancement, medical data processing, and VR-assisted interventions [17].

Recent studies suggest that VR and AI can be particularly effective in fields like radiology, where precise spatial understanding, image interpretation, and diagnostic skills are critical [18, 19]. The use of VR in radiology education is still in its early stages, but its potential for improving visualization and decision-making processes is evident [20]. AI's role in assisting radiologists by automating image analysis and offering real-time feedback could enhance the effectiveness of VR-based training platforms, allowing students to better grasp the intricacies of diagnostic procedures [21, 22]. Despite this potential, most research examines AI or VR in isolation, with limited studies exploring their integration to enhance diagnostic accuracy and clinical decision-making. Furthermore, strategies for effectively embedding these technologies into pedagogical frameworks remain underdeveloped.

To address this gap, the present study introduces VISTA-MED, an integrated AI-VR platform designed for real-time MRI visualization and segmentation in an immersive environment. This system aims to improve spatial understanding, diagnostic performance, and learner engagement. By investigating the synergistic potential of AI and VR, this research seeks to inform the development of innovative educational tools that bridge technology and healthcare training.

Methods

The methodology focuses on developing and validating the VISTA-MED system, an integrated platform combining Virtual Reality (VR) and Artificial Intelligence (AI) for real-time MRI visualization and

segmentation. The approach includes data acquisition and preprocessing, VR environment development, and AI model integration for automated segmentation.

Data Acquisition and Preprocessing

MRI datasets were obtained from publicly available repositories, including BRATS2020. All data were anonymized prior to processing. Preprocessing steps included bias field correction, image registration to the MNI152 template, and noise reduction using standard neuroimaging tools such as FSL and SPM. Regions of interest were segmented and verified for anatomical accuracy. Processed images were saved in NIfTI format to ensure compatibility with VR development workflows. Figure 1 shows Data preprocessing.

VR Environment Development

An interactive VR platform was developed using Unity to enable immersive visualization and manipulation of 3D MRI datasets. The environment supports intuitive interactions such as rotation, zooming, and region-specific exploration of anatomical structures. The conceptual frameworks guiding the development are primarily rooted in Cognitive Load Theory (CLT) '*suggests that working memory has limited capacity*', which can create a bottleneck when learning intricate medical concepts, especially in fields like anatomy [23], Constructivist Learning Theory, experiential learning where knowledge is constructed through interaction with the environment [24, 25], and the Technology Acceptance Model (TAM) provides a framework for understanding how users accept and adopt '*new technologies based on their perceptions of usefulness and ease of use*' [26] which collectively informs the integration of Virtual Reality (VR) and Artificial Intelligence (AI) into medical education. Educational modules were embedded to facilitate anatomy and imaging concepts. Internal testing confirmed rendering accuracy and stable frame rates for real-time interaction. Figure 2 shows the virtual reality environment.

Together, these theories provide a solid foundation for designing and evaluating VR and AI-based interventions aimed at enhancing medical education, particularly in the learning of anatomy and clinical diagnosis skills. By focusing on reducing cognitive load, promoting active learning, and ensuring the acceptance of technology by both students, educators, and professionals, the project seeks to develop effective, engaging, and impactful educational tools that align with contemporary pedagogical approaches.

Results

AI Integration for MRI Segmentation

Automated segmentation was implemented using the nnU-Net framework, selected for its adaptability and automated configuration for biomedical imaging tasks. The model was trained on the BRATS2020 dataset, which includes multi-modal MRI scans with annotated tumor regions. The model's

segmentation performance was measured using Dice, IoU, and precision–recall indicators. Figure 3 shows the workflow of model selection.

AI Model Performance

The nnU-Net framework was successfully implemented for automated segmentation of brain tumor regions using the BRATS2020 dataset. The model demonstrated strong performance across multiple evaluation metrics. The Dice coefficient achieved a mean score of 0.914, indicating high overlap between predicted segmentation masks and ground truth annotations. Intersection over Union (IoU) values were consistent with Dice scores, confirming accurate delineation of tumor boundaries. Precision-recall analysis revealed balanced performance, with minimal false positives and false negatives across segmentation thresholds. Figure 4 shows the methodology for the analysis in detail.

Images A and B in Figure 4 shows segmentation results for brain tumor regions were visualized alongside the input MRI modalities for qualitative evaluation. Graph D in Figure 4 represents training and validation metrics, observed reductions in training and validation losses across epochs demonstrate effective parameter tuning and model learning.. The convergence of these losses suggests good generalization. Graph E in Figure 4 indicates the learning rate of the model. A steadily decreasing learning rate, demonstrating a linear decay schedule. It ensures the model fine-tunes itself carefully as training progresses.

Training and Validation Metrics

Model training exhibited stable convergence, as evidenced by the consistent decline across training and validation over successive epochs. The learning rate followed a linear decay schedule, ensuring gradual fine-tuning during optimization. These trends suggest robust generalization and minimal overfitting.

Qualitative Evaluation

Segmentation outputs were visualized alongside input MRI modalities for qualitative assessment. Tumor regions were accurately identified and delineated, with clear separation from surrounding anatomical structures. Visual inspection confirmed that the AI-generated masks maintained anatomical fidelity, supporting their integration into immersive visualization workflows.

Discussion

The findings from this study underscore the transformative potential of integrating Virtual Reality (VR) and Artificial Intelligence (AI) in medical education. By combining immersive visualization with automated segmentation, the VISTA-MED platform addresses critical limitations of traditional anatomy and imaging instruction. Existing literature has consistently demonstrated the benefits of VR for enhancing spatial understanding and engagement (Zhao et al., 2020; Milykh et al., 2023), while AI-driven segmentation improves diagnostic precision and workflow efficiency (Najjar et al., 2023). Our approach

builds on these strengths by creating a unified system that enables real-time interaction with MRI data in a three-dimensional environment.

The integration of AI and VR offers several pedagogical advantages. First, it reduces cognitive load by presenting complex anatomical structures in an intuitive format, aligning with principles of Cognitive Load Theory. Second, it supports constructivist learning by allowing students to actively manipulate and explore anatomical models, fostering deeper comprehension. Third, the system's adaptability to diverse learning styles, including neurodiverse profiles positions it as an inclusive educational tool. These features collectively enhance diagnostic reasoning and spatial visualization, skills that are essential for radiology and other image-intensive disciplines.

Despite these promising outcomes, challenges remain. Technical requirements, such as high-performance hardware and VR headsets, may limit accessibility in resource-constrained settings. Additionally, while AI segmentation demonstrated strong accuracy in preliminary tests, further validation across diverse datasets and clinical scenarios is necessary.

Future Directions

Future evaluation will focus on usability, spatial understanding, and diagnostic accuracy through structured user studies. These assessments will employ validated instruments such as the System Usability Scale (SUS) and performance-based tasks to measure educational impact and clinical relevance. Future research aims to explore scalability, cost-effectiveness, and integration into existing curricula, as well as longitudinal studies to assess knowledge retention and clinical performance.

Conclusion

This study demonstrates that the convergence of AI and VR can significantly advance medical education by improving spatial understanding, diagnostic accuracy, and learner engagement. The VISTA-MED platform exemplifies how immersive technologies can bridge the gap between theoretical knowledge and hands on practice, equipping learners for complex diagnostic tasks in real-world clinical environments. By addressing current limitations in anatomy and imaging education, this research contributes to the growing body of evidence supporting technology-enhanced learning in healthcare.

Looking ahead, the integration of AI-VR systems into medical curricula has the potential to redefine training paradigms, making education more interactive, personalized, and clinically relevant. Continued refinement, broader implementation, and rigorous evaluation will be essential to realize this vision and ensure that future healthcare professionals are equipped with the skills required for precision medicine and advanced diagnostic practice.

Declarations

Ethics approval and consent to participate

This study was approved by Institutional Review Board (IRB) of Truman State University.

Consent for publication

All participants have consented to the academic publication.

Availability of data and materials

The datasets collected and/or analyzed during the current study are not publicly available but are available from the corresponding author, request.

Competing interests

The authors declare no competing interests.

Clinical Trials

Clinical trial number: not applicable.

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No funding was received for this study.

Authors' contributions

GS contributed to the conception of research idea and design, data collection, data analysis, and manuscript writing. MG contributed to the development of VR environment. PG contributed to data collection and data analysis, and manuscript writing. AR contributed to data collection and data analysis. All authors read and approved of the final manuscript.

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References

1. Arjun B. C., Gowda, S. (2024). Immersive Learning: Virtual Reality's Impact on Medical Curriculum. *International Scientific Journal of Engineering and Management*.
2. Salimi, S., Asgari, Z., Mohammadnejad, A., Teimazi, A., & Bakhtiari, M. (2024). Efficacy of virtual reality and augmented reality in anatomy education: A systematic review and meta-analysis. *Anatomical sciences education*.
3. Mistry, D., Brock, C.A., & Lindsey, T. (2023). The Present and Future of Virtual Reality in Medical Education: A Narrative Review. *Cureus*, 15.
4. Pottle, J. (2019). Virtual reality and the transformation of medical education. *Future Hospital Journal*, 6, 181 - 185.
5. Zhang, Y.Y., Feng, H., Zhao, Y., & Zhang, S. (2024). Exploring the Application of the Artificial-Intelligence-Integrated Platform 3D Slicer in Medical Imaging Education. *Diagnostics*, 14.
6. Naqvi, W.M., Sundus, H., Mishra, G.V., Muthukrishnan, R., & Kandakurti, P.K. (2024). AI in Medical Education Curriculum: The Future of Healthcare Learning. *European Journal of Therapeutics*.
7. Sun, L., Yin, C., Xu, Q., & Zhao, W. (2023). Artificial intelligence for healthcare and medical education: a systematic review. *American journal of translational research*, 15 7, 4820-4828.
8. Pujol, S., Baldwin, M., Nassiri, J.D., Kikinis, R., & Shaffer, K. (2016). Using 3D Modeling Techniques to Enhance Teaching of Difficult Anatomical Concepts. *Academic radiology*, 23 4, 507-16.
9. Triepels, C.P., Smeets, C.F., Notten, K.J., Kruitwagen, R.F., Futterer, J.J., Vergeldt, T.F., & van Kuijk, S.M. (2019). Does three-dimensional anatomy improve student understanding? *Clinical Anatomy (New York, N.y.)*, 33, 25 - 33.
10. Zhao, J., Xu, X., Jiang, H., & Ding, Y. (2020). The effectiveness of virtual reality-based technology on anatomy teaching: a meta-analysis of randomized controlled studies. *BMC Medical Education*, 20.
11. Milytkh, I.S., Kafarov, E.S., Covantsev, S., Dadashev, A.S., Skarlis, A.A., & Zenin, O.K. (2023). A new dimension in medical education: Virtual reality in anatomy during COVID-19 pandemic. *Clinical Anatomy*, 36, 1007 - 1015.
12. Jans, C., Bogossian, F.E., Andersen, P., & Levett-Jones, T. (2023). Examining the impact of virtual reality on clinical decision making - An integrative review. *Nurse education today*, 125, 105767.
13. King, F., Jayender, J., Bhagavatula, S.K., Shyn, P.B., Pieper, S.D., Kapur, T., Lasso, A., & Fichtinger, G. (2016). An Immersive Virtual Reality Environment for Diagnostic Imaging. *J. Medical Robotics Res.*, 1, 1640003:1-1640003:9.
14. Hattab, G., Hatzipanayioti, A., Klimova, A., Pfeiffer, M., Klausning, P., Breucha, M., Bechtolsheim, F.V., Helmert, J.R., Weitz, J., Pannasch, S., & Speidel, S. (2021). Investigating the utility of VR for spatial understanding in surgical planning: evaluation of head-mounted to desktop display. *Scientific Reports*, 11.
15. Prabha, N., Ziyad, N., Prasad, N., Abraham, J.P., Paul T, P., & T Paul, R. (2024). Enhanced Medical Analysis: Leveraging 3D Visualization and VR-AR Technology. *Journal of Sensor Networks and Data Communications*.

16. Sharki, J., Balushi, G.A., Maran, P., Ibrahim, M., Jabri, A.A., Palarimath, S., & Balakumar, C. (2024). Incorporating Artificial Intelligence Powered Immersive Realities to Improve Learning using Virtual Reality (VR) and Augmented Reality (AR) Technology. *2024 3rd International Conference on Applied Artificial Intelligence and Computing (ICAAIC)*, 760-765.
17. Wu, Y., Hu, K., Chen, D.Z., & Wu, J. (2024). AI-Enhanced Virtual Reality in Medicine: A Comprehensive Survey. *International Joint Conference on Artificial Intelligence*.
18. Najjar, R. (2023). Redefining Radiology: A Review of Artificial Intelligence Integration in Medical Imaging. *Diagnostics*, 13.
19. Lang, M., Ghandour, S., Rikard, B., Balasalle, E., Rouhezamin, M.R., Zhang, H., & Uppot, R.N. (2024). Medical Extended Reality for Radiology Education and Training. *Journal of the American College of Radiology : JACR*.
20. Xu, X., Mangina, E.E., Kilroy, D., Curran, K.M., Healy, J.J., & Campbell, A.G. (2020). Doctoral Colloquium—A Snapshot of the Future: Virtual and Augmented Reality Training for Radiology. *2020 6th International Conference of the Immersive Learning Research Network (iLRN)*, 407-410.
21. Ahmed, A., Rajarajeswari, S., Mallegowda, M., & Subash, N. (2024). The Power of Virtual Reality-Next-Gen Radiology Training. *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, 1-6.
22. Shetty, S.R., Bhat, S., Al Bayatti, S.W., Al Kawas, S., Talaat, W., El-Kishawi, M., Al Rawi, N., Narasimhan, S., Al-Daghestani, H., Madi, M., & Shetty, R. (2024). The Scope of Virtual Reality Simulators in Radiology Education: Systematic Literature Review. *JMIR Medical Education*, 10.
23. Young, J.Q., van Merrienboer, J.J., Durning, S.J., & ten Cate, O. (2014). Cognitive Load Theory: Implications for medical education: AMEE Guide No. 86. *Medical Teacher*, 36, 371 - 384.
24. Erawati, N.K., & Adnyana, P.B. (2024). Implementation of Jean Piaget's Theory of Constructivism in Learning: A Literature Review. *Indonesian Journal of Educational Development (IJED)*.
25. Kumari, G. (2022). Constructivism in Learning. *Educational Quest- An International Journal of Education and Applied Social Sciences*.
26. Einloft, J., Bedenbender, S., Michelsen, M., Meyer, H.L., Russ, P.G., Heidtmann, A., Hirsch, M.C., & Grgic, I. (2024). Structured Exposure Achieves High Acceptance of Immersive Technology Among Medical Students and Educators. *Cyberpsychology, behavior and social networking*.

Figures

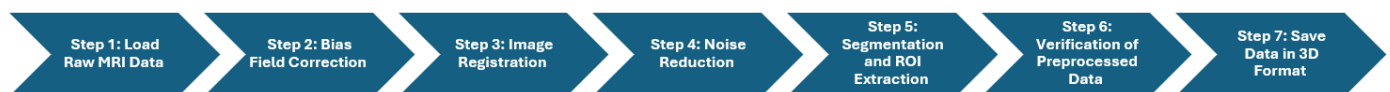


Figure 1

Data Preprocessing

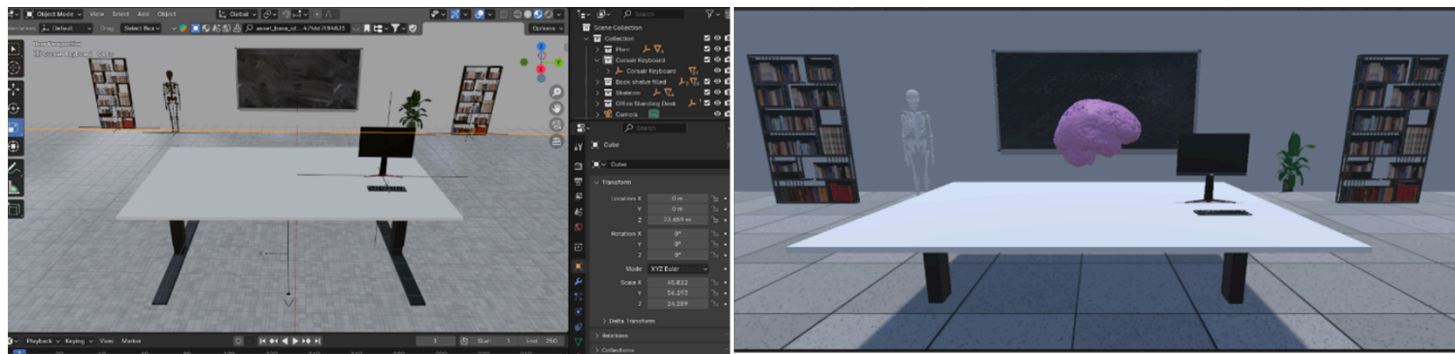


Figure 2

Sample VR Environments

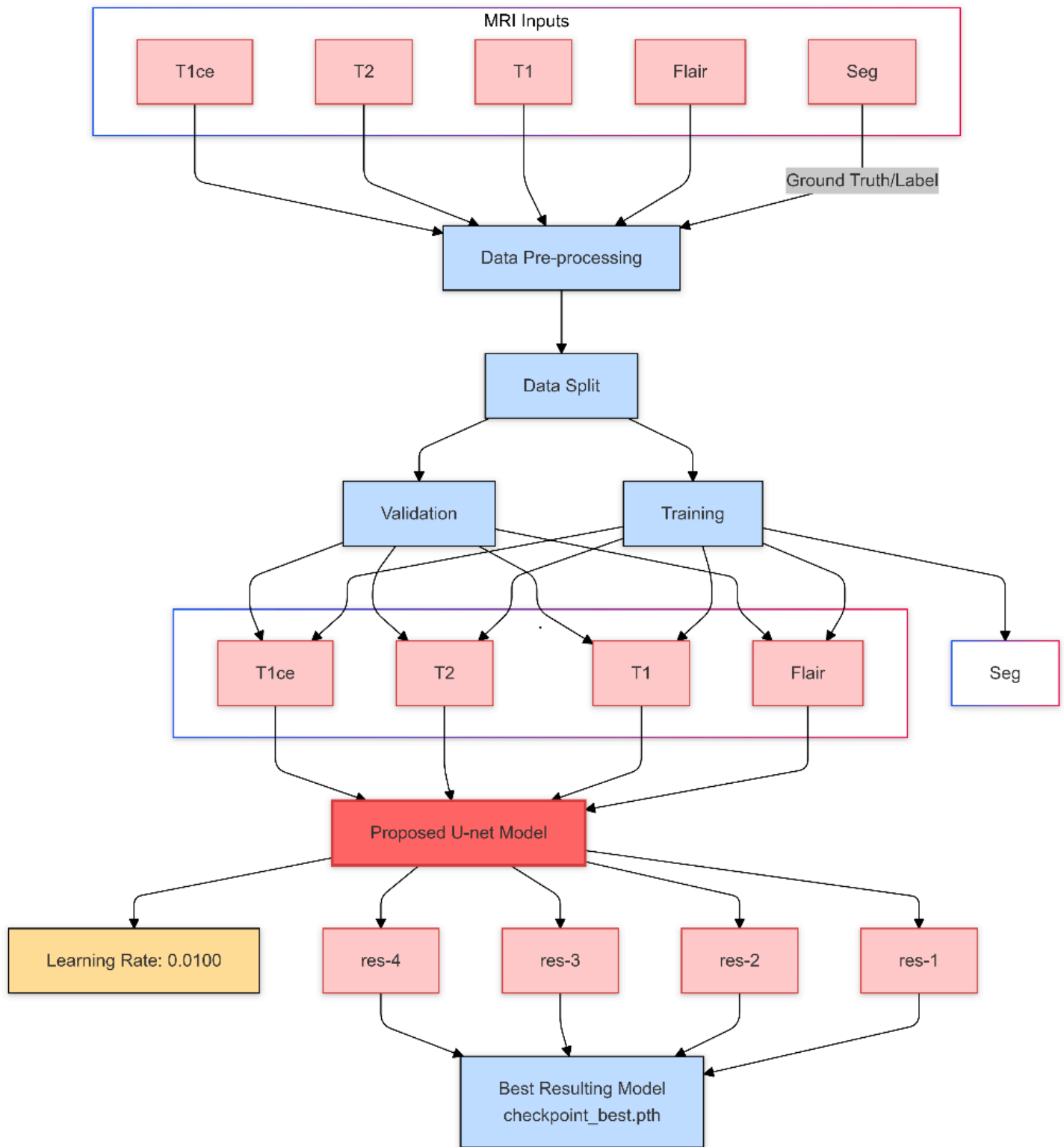


Figure 3

Pre-processing, training, and evaluation of MRI sequences for optimal model selection.

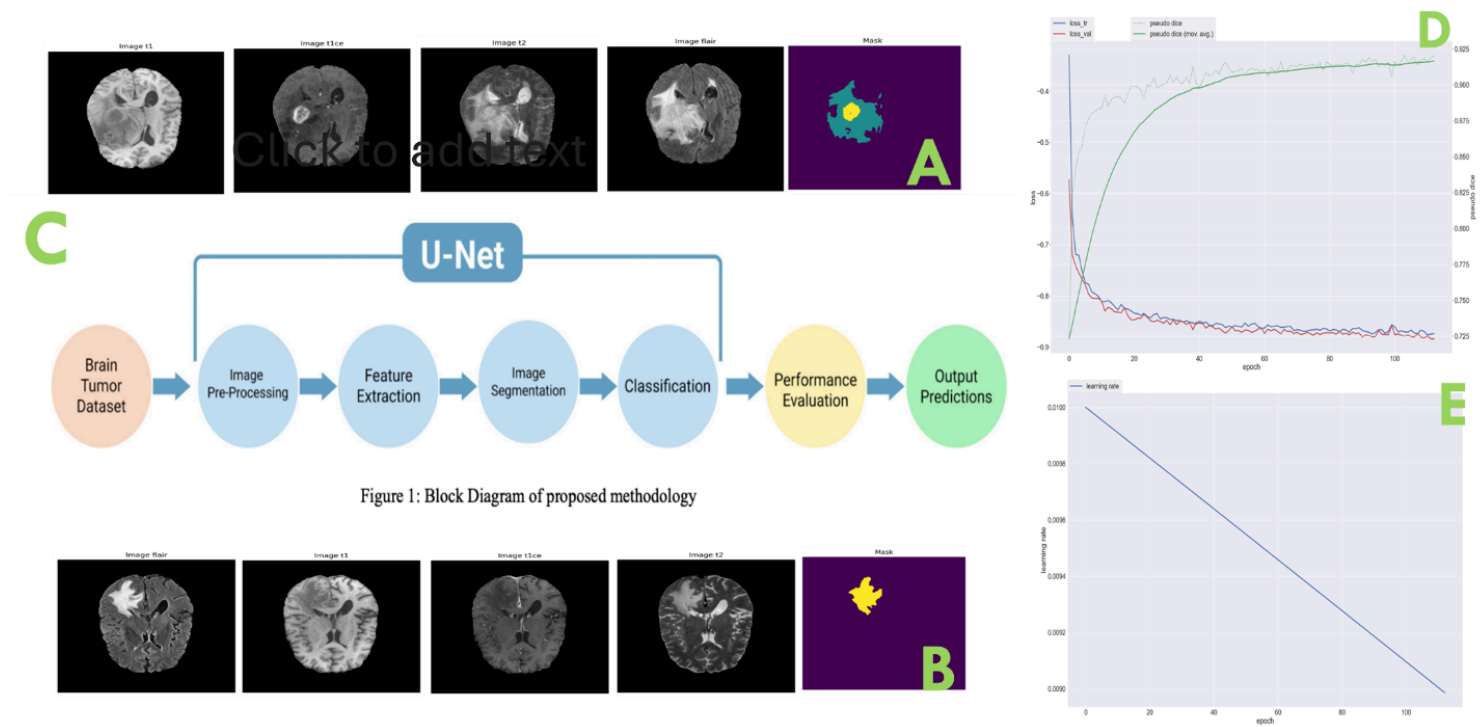


Figure 4

Observations from the Vizualized predictions